

EXECUTIVE INVESTMENT MEMORANDUM

PROJECT TITAN: 2-GIGAWATT AI DATA CENTER (TEXAS ERCOT REGION)

Model Architect: Qwen 2.5 Coder 14B Instruct | Benchmark Run: October 2024 | Classification: Tier-IV Mission Critical

This document provides an institutional investment analysis evaluating the capital deployment, civil engineering feasibility, power interconnection, and financial return profile for constructing a **2.0 Gigawatt AI hyperscale facility** in Texas. Based on the economic projections synthesized by **Qwen 2.5 Coder 14B Instruct**, total Capital Expenditure (CapEx) is modeled at **\$750,000,000** (Base Range: \$90,000,000 to \$1,500,000,000), with annualized Operational Expenditures (OpEx) of **\$65,000,000/year**. The facility is engineered to support ultra-dense AI clusters exceeding 100 kW/rack with redundant grid feeds.

Total Planned Capacity	Target Facility Area	Estimated CapEx (Mid)	Est. Annual OpEx	Target PUE
2.0 GW (2,000 MW)	500,000 sq ft	\$750,000,000	\$65,000,000	1.15 - 1.25

Capital Expenditure Allocation & Cost Centers

Item	Cost Center Description	Amount (\$USD)	Share (%)
1	Land & Civil Works	\$15,000,000	2.0%
2	Architectural, Structural & MEP Engineering	\$35,000,000	4.7%
3	High-Voltage Power Systems & Generators	\$350,000,000	46.7%
4	High-Density Chiller & Cooling Plants	\$180,000,000	24.0%
5	High-Speed Optical Fabric & Equipment	\$120,000,000	16.0%
6	Contingency & Financing Debt Reserve	\$50,000,000	6.7%
TOTAL CAPITAL INVESTMENT		\$750,000,000	100.0%

Strategic Architecture & Critical ERCOT Grid Risk Analysis

- 1. ERCOT Substation Interconnection:** Interconnecting a continuous 2.0 GW load requires direct 345 kV dual-circuit transmission lines, on-site utility substations, and multi-year coordination with ERCOT grid operators. Peak summer heat and grid curtailment necessitate on-site generation and battery backup.
- 2. Advanced Cooling Architecture:** With high-density AI accelerators exceeding 1 kW per socket, standard air cooling is insufficient. The model specifies closed-loop liquid-to-liquid heat exchangers and immersion manifolds, achieving an optimized PUE while conserving scarce regional water reserves.
- 3. Investment Recommendation:** The economics confirm a highly defensible infrastructure asset. Initial phasing in 250 MW to 500 MW data hall increments provides de-risked capital deployment and allows forward cash flows to fund sequential phases.